

System Architecture & Deployment Blueprint

This document outlines the core structural components and service dependencies of our next-generation cloud platform.

1. Component Specifications

- Frontend Web App: Built using React and Vite, serving as the user entrypoint for the dashboard interface.
- API Gateway: Handles request routing, rate limiting, and acts as the secure entry gateway.
- Auth Service: Validates user identity tokens and manages access permissions.
- Backend Engine: A scalable Node.js microservice executing core business algorithms and data retrieval.
- PostgreSQL Database: The primary persistent storage for system-wide structured tables.
- Redis Cache: Stores temporary keys and query responses to lower read latencies.
- Prometheus Service: Scrapes metrics across all running microservices for real-time alerting.

2. Service Interactions and Relationships

- The Frontend Web App sends client HTTPS requests to the API Gateway.
- The API Gateway proxies authentication requests directly to the Auth Service.
- The API Gateway routes application data requests directly to the Backend Engine.
- The Backend Engine retrieves and stores persistent state in the PostgreSQL Database.
- The Backend Engine caches hot queries inside the Redis Cache to bypass slow database calls.
- The Prometheus Service actively monitors the performance metrics of the Backend Engine.

Note: This blueprint is for Graph-RAG ingestion testing. All relations defined herein will be dynamically parsed and rendered.